Comfort Cuff: A Wearable System for Detecting and Making Menstrual Pain Visible

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Abstract

Menstrual pain (dysmenorrhea) and related gynecological conditions are common yet frequently normalized, under-reported, and under-diagnosed. The Comfort Cuff is a wearable system designed to detect abdominal cramping and associated pain using abdominal electromyography (EMG) together with complementary biosensors (heart rate variability, galvanic skin response, and motion). Users participate in a training period with button-press annotations, enabling personalized detection. Data are processed locally on a microcontroller and uploaded securely when possible to generate time-stamped logs for clinical evaluation, research, and advocacy. This paper reviews the biological basis of menstrual pain, wearable sensing strategies, design rationale, multimodal signal integration, implementation considerations, and societal impact. Recommendations for prototyping and future clinical evaluation are also discussed.

Keywords: dysmenorrhea, wearable sensors, EMG, HRV, GSR, pain detection, patient advocacy

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Introduction

Menstrual pain is highly prevalent, with conditions like primary dysmenorrhea, endometriosis, pelvic inflammatory disease, and polycystic ovary syndrome affecting millions worldwide (De Corte et al., 2025; MacGregor et al., 2023). Despite this, pain is often normalized or dismissed, delaying diagnosis and treatment. Marginalized populations—including women of color and low socioeconomic status—face systemic disparities in care, leading to untreated chronic pain (Wandner et al., 2012).

Comfort Cuff is proposed as a wearable robotic sensor system that detects uterine/abdominal cramping through abdominal EMG, complemented by HRV, GSR, motion, and user-annotated events. The device aims to: (1) provide patients with objective evidence to present to clinicians, (2) enable longitudinal research into treatments and triggers, and (3) increase social visibility of menstrual pain. Optional visibility features, including an LCD and buzzer, translate hidden pain into a public signal. This project integrates robotics (sensor integration, embedded processing), biomedical sensing, human factors, and healthcare data systems.

Biological Basis of Pain

Menstrual cramps result from prostaglandins, fatty acid–derived compounds that stimulate uterine contractions and promote inflammation. During menstruation, declining progesterone levels trigger a cascade in which phospholipase A2 releases arachidonic acid, which COX-1 and COX-2 convert into prostaglandin H2. PGH2 is then transformed into PGF2α, driving uterine contractions, and PGE2, amplifying inflammation and pain perception (Harel, 2006; MacGregor et al., 2023). These contractions reduce oxygen supply to uterine tissue, activating nociceptors and sending pain signals to the brain.

Muscle contractions generate electrical activity detectable with EMG. While most contractions are non-painful, strong uterine contractions produce signals that, combined with autonomic measures, indicate pain. Pain activates the sympathetic nervous system (“fight or flight”), increasing heart rate, reducing heart rate variability (HRV), and increasing skin conductance (GSR) via sweat gland activation (Loggia et al., 2011; Wang et al., 2016; Cascella et al., 2023).

Chronic or unmanaged menstrual pain can sensitize the nervous system (central sensitization), intensifying future cramps and producing pain in unrelated organs, known as “referred pain” (MacGregor et al., 2023). Endometriosis, in which endometrial-like tissue grows outside the uterus, can produce prostaglandin-mediated pain at ectopic sites. Pain from this tissue is not considered referred pain but underscores the need for multimodal sensing to capture non-uterine sources (Hellman et al., 2018).

Undiagnosed Conditions and Disparities

Despite the prevalence of menstrual pain, conditions like endometriosis, pelvic inflammatory disease, uterine fibroids, and PCOS may remain undiagnosed for years, sometimes over a decade, leaving individuals to manage debilitating pain with limited care (De Corte et al., 2025). Research shows women, racial and ethnic minorities, and elderly patients are more likely to have their pain underestimated or dismissed, leading to inequitable treatment (Wandner et al., 2012). Tools like Comfort Cuff can reduce these disparities by providing objective, shareable pain data for clinicians and researchers.

Detecting Pain Using Wearable Technology

Multimodal Sensing

Pain is multidimensional, encompassing physiological, behavioral, and subjective elements. Comfort Cuff combines:

- **Abdominal EMG:** Captures uterine contractions; sensitive to motion and electrode placement (Clancy, Morin, & Merletti, 2002).
- **Heart rate / HRV:** Pain-induced sympathetic activation reduces HRV (Wang et al., 2016).
- **Galvanic skin response (GSR):** Sympathetic activation increases skin conductance (Cascella et al., 2023).
- **Motion / accelerometry:** Detects posture changes indicative of pain, reducing false positives (Cascella et al., 2023).
- **User annotations:** Button presses during pain events provide ground truth for model calibration.

Signal Quality and Preprocessing

EMG is prone to noise; mitigation includes proper electrode choice, skin preparation, shielding, and digital preprocessing such as filtering and smoothing (Clancy et al., 2002). Dry electrodes in textiles can approximate wet electrode performance while improving comfort (Angelucci et al., 2021).

Prior Wearables and Gaps

Wrist-based wearables like Empatica E4 measure PPG, GSR, motion, and temperature, and can classify menstrual phase with moderate accuracy (Kilungeja et al., 2025; Cascella et al., 2023). These devices, however, lack abdominal EMG and do not provide clinician-ready cramp detection, leaving Comfort Cuff to fill this gap.

Multi-Population Applications

- **Patients / Advocacy:** Converts subjective pain into time-stamped, shareable logs to reduce dismissal and enable accurate diagnosis (De Corte et al., 2025; Wandner et al., 2012).
- **Clinical Research:** Supports longitudinal evaluation of interventions (MacGregor et al., 2023).
- **Public Awareness:** Optional visibility mode externalizes hidden pain, promoting societal recognition of menstrual pain.

Design and Development

High-Level Overview

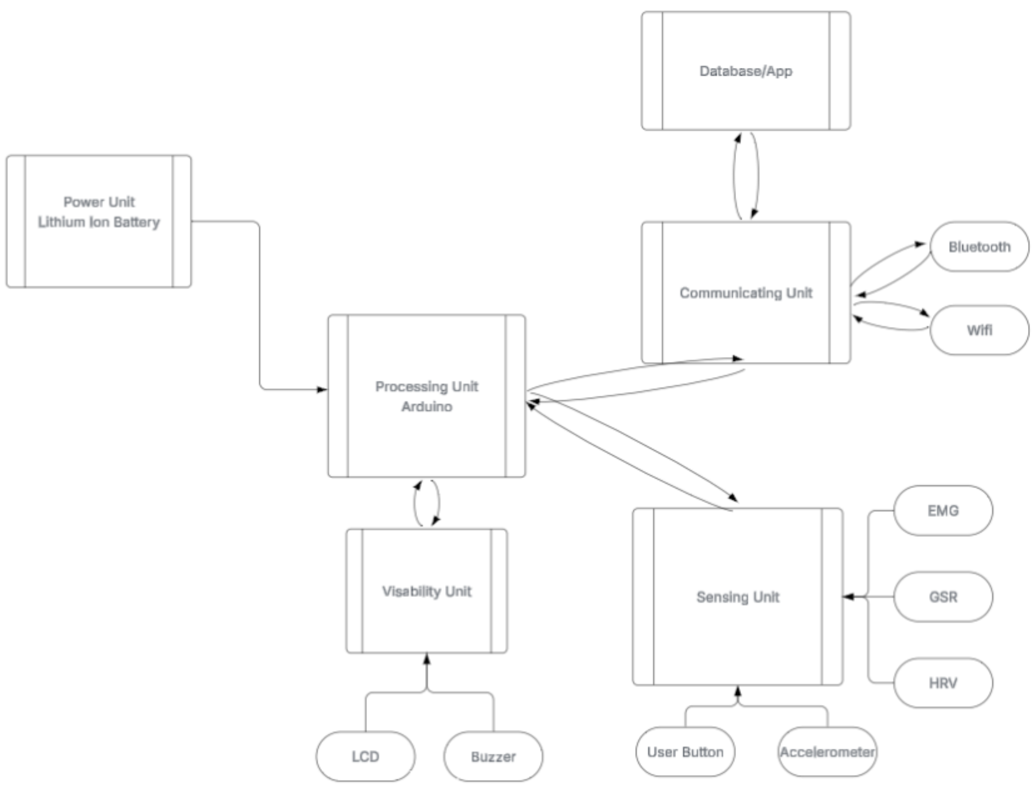

Fig 1. Block Diagram of Comfort Cuff's components

- **Sensors:** Abdominal EMG, HRV/PPG, GSR, accelerometer, push-button annotation.
- **Processing / Microcontroller:** Initially Arduino for prototyping; ESP32 or similar recommended for wireless/low-power future iterations.
- **Connectivity:** Wi-Fi or Bluetooth-to-phone for secure upload to cloud database; optional clinician/research dashboards.

- **User Interface:** Minimal device UI; optional LCD/buzzer visibility mode; companion app for visualization and export.
- **Power:** Rechargeable lithium-ion battery supporting ~12 hours of continuous monitoring.

Logic / Data Flow

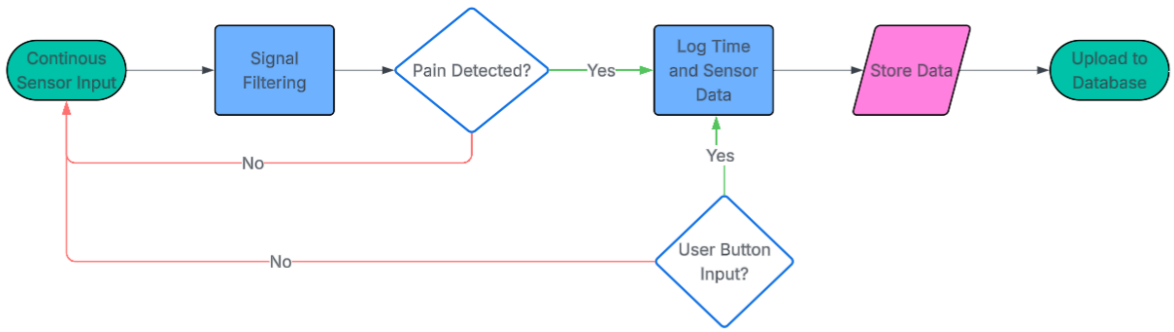

Fig 2. Diagram depicting logic flow of Comfort Cuff

1. Continuous sensor sampling at appropriate rates.
2. Event annotation with button press.
3. Preprocessing: filtering, smoothing, feature extraction (EMG amplitude/rate, HRV features, GSR phasic changes, accelerometry).
4. On-device thresholding/classification based on training period.
5. Local storage and asynchronous, encrypted cloud upload.
6. Visualization for patients and clinicians to analyze trends over time (Feller et al., 2018).

Design Decisions

- EMG for proximal uterine signal (Hellman et al., 2018).
- Multimodal sensing to improve specificity and reduce false positives (Wang et al., 2016; Cascella et al., 2023).
- User training for personalization.
- Smart textile dry electrodes for comfort and adherence (Angelucci et al., 2021).

Implementation Considerations

- **Signal quality & motion artifacts:** Noise mitigation and preprocessing needed.
- **False negatives for non-uterine pain:** Some endometriosis patients may lack expected EMG signatures (Hellman et al., 2018).
- **Generalization:** Individual differences necessitate user-specific calibration (Cascella et al., 2023).
- **Socio-technical risks:** Adoption barriers, privacy concerns, and accessibility issues must be addressed (Greenhalgh et al., 2017; Smuck et al., 2021).

Medical Device Success and Existing Gaps

Successful clinical wearables require clear problem definition, user-friendly design, and clinician integration (Smuck et al., 2021). Comfort Cuff meets these criteria while addressing gaps in existing devices (Empatica E4, DaVinci wearable) by combining EMG-based cramp detection, multimodal sensors, annotation-driven personalization, and clinician-ready exports.

Conclusion

Comfort Cuff provides a multimodal, patient-centered wearable solution to menstrual pain detection. By combining abdominal EMG with autonomic and behavioral sensors, user-driven training, and secure data management, it can reduce clinical dismissal, support research, and increase societal awareness. Next steps include prototyping, signal validation, user-centered testing, and engagement with clinical partners. Investments in devices like Comfort Cuff represent a tangible opportunity to improve equity, visibility, and outcomes in women's health.

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